

1 Introduction

Notation:

- Random variable X_i with value x_i and possible values from $\mathcal{X}$,
- Random vector $\mathbf{X}$ with range $\mathcal{X}$ and some assignment $\mathbf{x}$,
- Set $\mathcal{A}$, and a measure $\mathbb{M}$.

2 Central Distribution

Definition 2.1 (Probabilistic Model). We define a *probabilistic model* $\mathcal{M}$ as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of probability measures on $\mathcal{S}$.

Definition 2.2 (Divergence Measure). Let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of pairs of discrete random variables in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *divergence measure* of $P \in \mathcal{P}$ for $\mathcal{X}$ as follows:

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}(Q_{X_i|e_i} \parallel P_{X_i|e_i}) \right], \quad (1)$$

where $\mathbb{E}_Q[\cdot]$ denotes expectation relative to Q and each $\mathbb{D}(Q_{X_i|e_i} \parallel P_{X_i|e_i})$ is the Kullback-Leibler divergence between $P(X_i|e_i)$ and $Q(X_i|e_i)$:

$$\mathbb{D}(Q_{X_i|e_i} \parallel P_{X_i|e_i}) = \sum_{x_i} Q(x_i | e_i) \ln \frac{Q(x_i | e_i)}{P(x_i | e_i)}.$$

We also define *the central distribution for $\mathcal{X}$* to be a distribution $C_{\mathcal{X}} \in \mathcal{P}$ which minimizes the divergence measure for $\mathcal{X}$:

$$C_{\mathcal{X}} = \arg \min_{P \in \mathcal{P}} \mathbb{D}_{\mathcal{X}}[P]. \quad (2)$$

Proposition. Using the linearity of expectation, we have

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{Q \in \mathcal{P}} \sum_i \sum_{x_i, e_i} Q(x_i, e_i) \ln \frac{Q(x_i | e_i)}{P(x_i | e_i)}.$$

Definition 2.3 (Parametric Model). Let Θ be a random vector. A parametric model is a probabilistic model $\mathcal{M}_{\Theta} \equiv (\mathcal{S}, \mathcal{P})$, where for every $P \in \mathcal{P}$ there is a prior probability distribution p , such that for every $s \in \mathcal{S}$ we have

$$P(s) = \int \pi(s | \theta) p(\theta) d\theta, \quad (3)$$

while π is a known probability distribution.

Lemma 1. Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of discrete random variable pairs in $\mathcal{M}_\Theta$. For every $P \in \mathcal{P}$ we have

$$\begin{aligned} \mathbb{D}_{\mathcal{X}}[P] &\leq \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right] \\ &= \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)}. \end{aligned}$$

Proof. Let R be an arbitrary distribution in $\mathcal{M}_\Theta$, since the Kullback-Leibler divergence is positive, its expected value is also positive, and for every parameter value θ we have

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{R(x_i \mid e_i)} \geq 0.$$

Using this, for every distribution P and every parameter θ , we obtain

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)}.$$

Since this inequality holds for any θ , the maximum of the right-hand side is also greater than or equal to the maximum of the left-hand side w.r.t. θ :

$$\max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right]. \quad (4)$$

$\mathcal{M}_\Theta$ is a parametric model and $R(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r(\theta) d\theta$, where $r(\theta)$ is a probability distribution. Clearly, for every θ we have

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

Since r is a distribution and is non-negative, we can multiply this inequality by $r(\theta)$ and integrate. Notice that the right hand side is not a function of θ and it will remain unchanged because $\int r(\theta) d\theta = 1$:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right] \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

Now from Inequality 4, for *any* distribution P and R in $\mathcal{M}_\Theta$, it follows that

$$\mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right] \leq \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right],$$

which proves the theorem. $\square$

Theorem 2. Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model and $\mathcal{X}$ be a set of pairs of discrete random variables in $\mathcal{M}_\Theta$. If for every value of θ there is a distribution $R \in \mathcal{P}$ such that $R(s) = \pi(s \mid \theta)$ for every $s \in \mathcal{S}$, then for any $P \in \mathcal{P}$ we have:

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right]. \quad (5)$$

Proof. For an arbitrary distribution $P \in \mathcal{P}$, we let

$$\theta^* = \arg \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right].$$

In $\mathcal{M}_\Theta$ there should be a distribution $R \in \mathcal{P}$ that equals $\pi(\cdot \mid \theta^*)$, and we can write

$$\mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right] = \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta^*} \parallel P_{X_i|e_i}) \mid \theta^* \right].$$

On the other hand, we see

$$\max_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}(Q_{X_i|e_i} \parallel P_{X_i|e_i}) \right] \geq \mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right].$$

Thus,

$$\mathbb{D}_{\mathcal{X}}[P] \geq \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right].$$

This inequality and Lemma 1 prove the theorem. $\square$

Theorem 3 (Minimax theorem [1]). Let $\mathcal{X} \subset \mathbb{R}^n$ and $\mathcal{Y} \subset \mathbb{R}^m$ be compact convex sets. If $f : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ is a continuous function that is concave-convex, i.e.

- $f(\cdot, y) : \mathcal{X} \rightarrow \mathbb{R}$ is concave for fixed y , and
- $f(x, \cdot) : \mathcal{Y} \rightarrow \mathbb{R}$ is convex for fixed x .

Then we have that

$$\max_{x \in \mathcal{X}} \min_{y \in \mathcal{Y}} f(x, y) = \min_{y \in \mathcal{Y}} \max_{x \in \mathcal{X}} f(x, y). \quad (6)$$

Theorem 4. Consider a generic parametric model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$, and let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of pairs of discrete random variables in $\mathcal{M}_\Theta$. Assume that for every $Q_1, Q_2 \in \mathcal{P}$ and any real number $0 \leq \alpha \leq 1$, there is a distribution $Q^* \in \mathcal{P}$, such that for every i, x_i, e_i we have

$$Q^*(x_i | e_i) = \alpha Q_1(x_i | e_i) + (1 - \alpha) Q_2(x_i | e_i). \quad (7)$$

In this model, the central distribution for $\mathcal{X}$ can be obtained by:

$$C_{\mathcal{X}} = \arg \max_{P \in \mathcal{P}} \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right] p(\theta) d\theta \quad (8)$$

$$= \arg \max_{P \in \mathcal{P}} \int \sum_i \sum_{x_i, e_i} P(x_i, e_i, \theta) \ln \frac{\pi(x_i | e_i, \theta)}{P(x_i | e_i)} d\theta, \quad (9)$$

where $P(x_i, e_i, \theta) = \pi(x_i, e_i | \theta) p(\theta)$.

Proof. We prove this theorem using the Minimax theorem. For that, we define a functional as follows:

$$f(r, P_{X_i|e_i}) = \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right] r(\theta) d\theta,$$

where $r(\theta)$ is an arbitrary probability distribution, and $P \in \mathcal{P}$. Since $r(\theta)$ can be any distribution, maximizing with respect to r is like maximizing with respect to θ , and we have

$$C_{\mathcal{X}} = \arg \min_{P \in \mathcal{P}} \max_r f(r, P_{X_i|e_i}).$$

The domain of r is the set of all probability distributions, which is a compact convex set. Moreover, from Equation 7, it follows that the set of probability distributions for $X_i|E_i$ in $\mathcal{M}_\Theta$ is also a compact convex set. Thus, the domain of $f(r, P_{X_i|e_i})$ is a compact convex set.

For a fixed $P_{X_i|e_i}$, $f(\cdot, P_{X_i|e_i})$ is a linear function, which is both concave and convex. On the other hand, if we let $R(x_i, e_i) = \int \pi(x_i, e_i | \theta) r(\theta) d\theta$ for $f(r, \cdot)$ we have

$$f(r, \cdot) = c - \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i | e_i). \quad (10)$$

If we calculate the Hessian matrix of $f(r, \cdot)$, we observe that the matrix is diagonal, and each element of the main diagonal is $\frac{R(x_i|e_i)}{P(x_i|e_i)^2}$. Thus, the hessian matrix of $f(\cdot, P_{X_i|e_i})$ is positive semi-definite and $f(r, \cdot)$ is a convex function. Now, Theorem 3 implies

$$\min_{P \in \mathcal{P}} \max_r f(r, P_{X_i|e_i}) = \max_r \min_{P \in \mathcal{P}} f(r, P_{X_i|e_i}). \quad (11)$$

From Equation 10 we can see that for an arbitrary fixed r we have

$$\min_{P \in \mathcal{P}} f(r, P_{X_i|e_i}) = \max_{P \in \mathcal{P}} \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i | e_i).$$

We know that the Kullback–Leibler divergence, and therefore its expected value, is non-negative:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right] \geq 0.$$

By simple algebra we get

$$\sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i | e_i) \leq \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln R(x_i | e_i).$$

Recall that $\mathcal{M}_\Theta$ is a generic parametric model, so $R \in \mathcal{P}$, and we can reach this upper bound by letting $P = R$:

$$\min_{P \in \mathcal{P}} f(r, P_{X_i|e_i}) = \int \sum_i \sum_{x_i, e_i} R(x_i, e_i, \theta) \ln \frac{\pi(x_i | e_i, \theta)}{R(x_i | e_i)} d\theta.$$

By Equation 11 we conclude

$$C_{\mathcal{X}} = \arg \max_{R \in \mathcal{R}} \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel R_{X_i|e_i}) \mid \theta \right] r(\theta) d\theta. \quad \square$$

Theorem 5. *Under the assumptions of Theorem 4, the central distribution for $\mathcal{X}$ satisfies:*

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i | \theta) \ln \frac{\pi(x_i | e_i, \theta)}{C_{\mathcal{X}}(x_i | e_i)} = \lambda,$$

where λ is a constant.

Proof. Since $\mathcal{M}_\Theta$ is a parametric model, the central distribution for $\mathcal{X}$ maximizes the right-hand side of Equation 9 with respect to different choices of the prior distribution $p(\theta)$, subject to the constraint $\int p(\theta) d\theta = 1$. Applying the Lagrange multiplier technique, the local maximum of the right-hand side of Equation 9 will be a stationary point of the Lagrangian function:

$$\mathcal{L}[p(\theta), \lambda] = \int \sum_i \sum_{x_i, e_i} P(x_i, e_i, \theta) \ln \frac{\pi(x_i | e_i, \theta)}{P(x_i | e_i)} d\theta - \lambda \left(\int p(\theta) d\theta - 1 \right).$$

Let us denote the prior distribution corresponding to $C_{\mathcal{X}}$ by $c(\theta)$. Let $p(\theta) = c(\theta) + \epsilon \eta(\theta)$ be a variation of $c(\theta)$, where $\eta(\theta)$ is a continuously differentiable function with $\eta(0) = \eta(1) = 0$, and ϵ is a small constant.

Since $c(\theta)$ is an extremizing function for the right-hand side of Equation 9, $\mathcal{L}[\epsilon, \lambda]$ has a stationary point where $\epsilon = 0$, and we have

$$\left. \frac{\partial \mathcal{L}}{\partial \epsilon} \right|_{\epsilon=0} = 0.$$

Note that

$$\begin{aligned} \frac{dp(\theta)}{d\epsilon} &= \eta(\theta), \\ \frac{dP(x_i, e_i)}{d\epsilon} &= \int \pi(x_i, e_i | \theta) \eta(\theta) d\theta, \\ \frac{dP(e_i)}{d\epsilon} &= \int \pi(e_i | \theta) \eta(\theta) d\theta. \end{aligned}$$

Using the chain rule, we calculate the partial derivative of $\mathcal{L}[\epsilon, \lambda]$ relative to ϵ :

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \epsilon} &= \int \left(\sum_i \sum_{x_i, e_i} \pi(x_i, e_i | \theta) \ln \frac{\pi(x_i | e_i, \theta)}{P(x_i | e_i)} \right) \eta(\theta) d\theta \\ &\quad - \int \left(\sum_i \sum_{x_i, e_i} \pi(x_i, e_i | \theta) \frac{\int \pi(x_i, e_i | \theta) \eta(\theta) d\theta}{P(x_i, e_i)} \right) p(\theta) d\theta \\ &\quad + \int \left(\sum_i \sum_{e_i} \pi(e_i | \theta) \frac{\int \pi(e_i | \theta) \eta(\theta) d\theta}{P(e_i)} \right) p(\theta) d\theta \\ &\quad - \int \lambda \eta(\theta) d\theta \end{aligned}$$

Since $P(x_i, e_i) = \int \pi(x_i, e_i | \theta) p(\theta) d\theta$ and $P(e_i) = \int \pi(e_i | \theta) p(\theta) d\theta$ we get

$$\frac{\partial \mathcal{L}}{\partial \epsilon} = \int \left(-\lambda + \sum_i \sum_{x_i, e_i} \pi(x_i, e_i | \theta) \ln \frac{\pi(x_i | e_i, \theta)}{P(x_i | e_i)} \right) \eta(\theta) d\theta.$$

We let $\left. \frac{\partial \mathcal{L}}{\partial \epsilon} \right|_{\epsilon=0} = 0$. Since $\eta(\theta)$ is an arbitrary function, by applying the fundamental lemma of the calculus of variations and letting $\epsilon = 0$, for every value of θ we have

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i | \theta) \ln \frac{\pi(x_i | e_i, \theta)}{C_{\mathcal{X}}(x_i | e_i)} = \lambda. \quad \square$$

Theorem 6. Let $\mathcal{M}_{\Theta} \equiv (\mathcal{S}, \mathcal{P})$ be a generic parametric model, and $\mathcal{X} = \{X_1, X_2, X_3, \dots\}$ be a set of random variables in $\mathcal{M}_{\Theta}$. Let for every i , Ω_i be a function of the parameter Θ that fully parametrizes X_i :

$$\pi(x_i | \theta) = \pi(x_i | \Omega_i(\theta) = \omega_i). \quad (12)$$

The prior distribution corresponding to the central distribution for $\mathcal{X}$ in $\mathcal{M}_\Theta$ can be characterized by the following fixed point equation:

$$c_{\mathcal{X}}(\theta) \propto \left(\prod_i C_{\mathcal{X}}(\theta | \omega_i) \rho(\omega_i) \right)^{\frac{1}{|\mathcal{X}|}}, \quad (13)$$

where

$$\rho(\omega_i) \propto \exp \left\{ \sum_{x_i} \pi(x_i | \omega_i) \ln C_{\mathcal{X}}(\omega_i | x_i) \right\}.$$

Proof. First, we show that the set of distributions of each X_i is a convex set. Let $Q^*(x_i) = \alpha Q_1(x_i) + (1 - \alpha) Q_2(x_i)$, for some $Q_1, Q_2 \in \mathcal{P}$, $0 \leq \alpha \leq 1$ and arbitrary i, x_i .

$\mathcal{M}_\Theta$ is a parametric model. Therefore,

$$\begin{aligned} Q^*(x_i) &= \alpha \int \pi(x_i | \theta) q_1(\theta) d\theta + (1 - \alpha) \int \pi(x_i | \theta) q_2(\theta) d\theta \\ &= \int \pi(x_i | \theta) (\alpha q_1(\theta) + (1 - \alpha) q_2(\theta)) d\theta. \end{aligned}$$

$\alpha q_1(x_i) + (1 - \alpha) q_2(x_i)$ is a valid prior distribution and $\mathcal{M}_\Theta$ is a generic parametric model. That implies $Q^* \in \mathcal{P}$.

Thus, by Theorem 5 the central distribution of $\mathcal{X}$ satisfies

$$\sum_i \sum_{x_i} \pi(x_i | \theta) \ln \frac{\pi(x_i | \theta)}{C_{\mathcal{X}}(x_i)} = \lambda. \quad (14)$$

Moreover, from Equation 12 and Bayes' theorem it follows

$$\begin{aligned} \sum_i \sum_{x_i} \pi(x_i | \theta) \ln \frac{\pi(x_i | \theta)}{C_{\mathcal{X}}(x_i)} &= \sum_i \sum_{x_i} \pi(x_i | \omega_i) \ln \frac{\pi(x_i | \omega_i)}{C_{\mathcal{X}}(x_i)} \\ &= \sum_i \sum_{x_i} \pi(x_i | \omega_i) \ln \frac{C_{\mathcal{X}}(\omega_i | x_i)}{c_{\mathcal{X}}(\omega_i)} \\ &= \sum_i \ln \frac{\rho(\omega_i)}{c_{\mathcal{X}}(\omega_i)}. \end{aligned}$$

Since each Ω_i is a function of Θ , we have $C_{\mathcal{X}}(\omega_i, \theta) = C_{\mathcal{X}}(\theta)$ and

$$c_{\mathcal{X}}(\omega_i) = \frac{c_{\mathcal{X}}(\theta)}{C_{\mathcal{X}}(\theta | \omega_i)}.$$

Now we rewrite Equation 14:

$$|\mathcal{X}| \ln c_{\mathcal{X}}(\theta) = \sum_i \ln (C_{\mathcal{X}}(\theta | \omega_i) \rho(\omega_i)) - \lambda$$

$$c_{\mathcal{X}}(\theta) = \left(e^{-\lambda} \prod_i C_{\mathcal{X}}(\theta | \omega_i) \rho(\omega_i) \right)^{\frac{1}{|\mathcal{X}|}} \quad \square$$

References

- [1] Ding-Zhu Du. *Minimax and Its Applications*, pages 339–367. Springer US, Boston, MA, 1995.